Project Proposal
IoT-Based Smart Patient Turning and Bed Monitoring System for Bedridden Patients

Course title: ***IOT Laboratory***
Course code: CSE-460
4th Year 1st Semester Examination 2025

Date of Submission: 26/04/2026

Submitted to-

Dr. Md. Ezharul Islam, B.SC.(JU), M.ENGG.(JAPAN), PHD(JAPAN)
Professor
Department of Computer Science and Engineering
Jahangirnagar University

&

Samsun Nahar Khandakar
Assistant Professor
Department of Computer Science and Engineering
Jahangirnagar University Savar,
Dhaka-1342

Class Roll	Exam Roll	Name
383	210903	Abdullah Nazmus-Sakib
379	210899	Shah Sultan
391	210911	Md Khaled Mahmud

Contents

Abstract

Bedridden patients are highly vulnerable to pressure ulcers (bed sores) due to prolonged immobility. Continuous manual monitoring is often inefficient and prone to delay.

This project proposes an IoT-Based Smart Patient Turning and Bed Monitoring System that continuously monitors patient body pressure, temperature, and humidity conditions. The system detects prolonged immobility and generates alerts to prevent pressure ulcers.

The system uses an ESP32 microcontroller integrated with pressure sensors and DHT22 environmental sensors. Data is transmitted to a cloud platform for real-time remote monitoring through a mobile application. Additionally, a buzzer provides local alerts.

Chapter 1: Introduction

In modern healthcare systems, bedridden patients require continuous monitoring to prevent complications such as pressure ulcers, muscle stiffness, and infections.

Manual monitoring by caregivers is time-consuming and may lead to delays in patient repositioning.

To address this issue, an IoT-based automated monitoring system is required to ensure continuous tracking and timely alerts.

Chapter 2: Literature Review

Several IoT-based healthcare monitoring systems have been developed in recent years, including:

- Heart rate monitoring systems
- Temperature monitoring devices
- Smart hospital beds

However, most existing systems do not integrate pressure detection with time-based position monitoring and automated alert generation in a single platform.

This project aims to fill this gap by combining multiple monitoring parameters into one integrated system.

🧾 Chapter 3: Problem Statement

The major problems in current patient care systems include:

- Lack of continuous monitoring for bedridden patients
- Delayed detection of pressure ulcers
- High dependency on manual caregiver observation
- Absence of real-time remote monitoring systems

🎯 Chapter 4: Objectives

The main objectives of this project are:

- To design a pressure monitoring system for bedridden patients
- To track patient immobility duration
- To develop an IoT-based alert system
- To send real-time notifications to caregivers
- To implement cloud-based data storage

🧾 Chapter 5: System Overview

The proposed system consists of four main components:

1. Sensor Unit
2. Processing Unit (ESP32 Microcontroller)
3. IoT Cloud Platform
4. Alert System (Buzzer + Mobile Notification)

The system continuously monitors patient condition and provides alerts when abnormal conditions are detected.

⚙ Chapter 6: Methodology

Step 1: Data Collection

- Pressure sensors (FSR or Load Cell) detect body pressure distribution
- DHT22 sensor measures temperature and humidity

Step 2: Data Processing

- ESP32 collects and processes sensor data
- System tracks duration of patient immobility

Step 3: Decision Making

- If the patient remains in the same position for a defined time (e.g., 2–3 hours), an alert is triggered

Step 4: Alert Generation

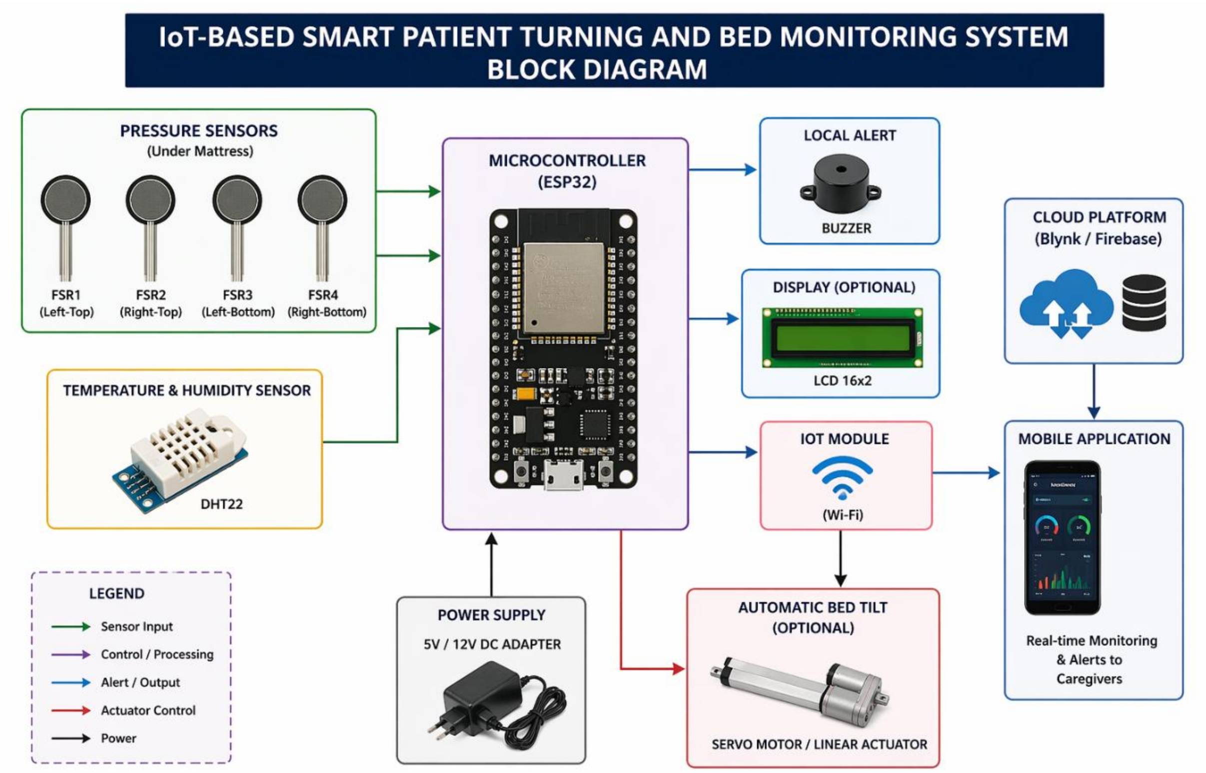
- Buzzer activates locally
- Mobile application sends notification to caregivers

Step 5: Cloud Integration

- Data is uploaded to Firebase or Blynk cloud
- Remote monitoring is enabled

🏗 Chapter 7: System Design

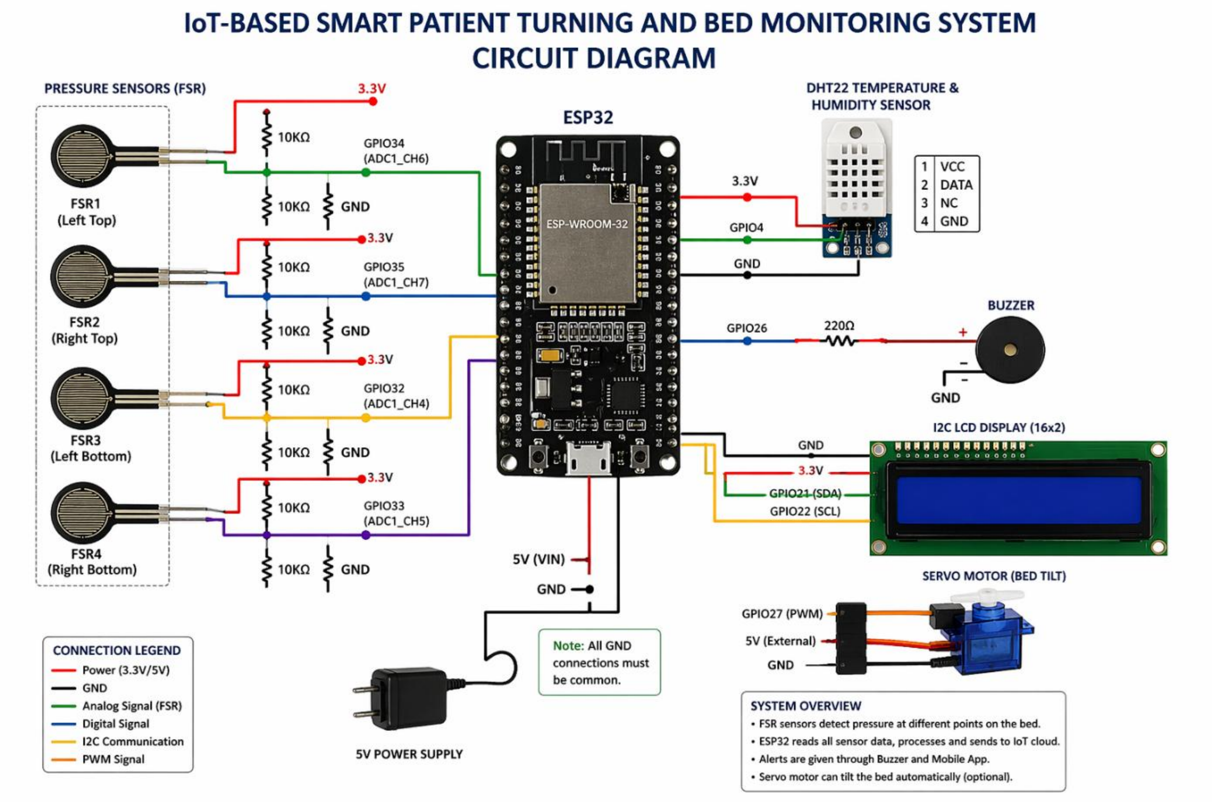
◇ 7.1 Block Diagram



◇ 7.2 Circuit Diagram

Hardware Connections:

- FSR Sensor → ESP32 GPIO
- DHT22 Sensor → Digital Input Pin
- Buzzer → GPIO Output Pin
- ESP32 → 5V Power Supply



🧪 Chapter 8: Implementation Plan

Hardware Components:

- ESP32 Microcontroller
- FSR / Load Cell Sensors
- HX711 Amplifier Module
- DHT22 Sensor
- Buzzer
- Power Supply

Software Tools:

- Arduino IDE
- Firebase / Blynk IoT Platform
- Mobile Application Dashboard

🧪 Chapter 9: Simulation Plan

The system will be simulated using:

- Wokwi Online ESP32 Simulator
- Proteus (optional hardware simulation)

Simulation Features:

- Pressure sensor input simulation
- Temperature and humidity monitoring
- Alert triggering logic testing
- IoT data transmission simulation

🎯 Chapter 10: Expected Outcome

The expected outcomes of this project include:

- Reduction of pressure ulcer risk
- Improved patient safety and comfort
- Real-time monitoring of patient condition
- Reduced workload for caregivers
- Affordable smart healthcare solution

⚠ Chapter 11: Limitations

- Requires stable internet connection
- Sensor calibration may be required
- Hardware accuracy limitations
- Power dependency for continuous operation

🚀 Chapter 12: Future Scope

Future enhancements may include:

- AI-based patient movement prediction
- Automatic bed repositioning system
- Integration with hospital management systems
- Voice alert system for caregivers
- Machine learning-based health prediction

Chapter 13: Cost Analysis (Bill of Materials)

This chapter presents the estimated cost of all hardware and components used in the proposed IoT-Based Smart Patient Turning and Bed Monitoring System.

📊 Table: Estimated Cost of Components

Sl. No	Component	Quantity	Approx. Cost (BDT)	Approx. Cost (USD)
1	ESP32 Development Board	1	600 – 900 BDT	5 – 8 USD
2	FSR Pressure Sensor / Load Cell	4	800 – 1500 BDT	7 – 12 USD
3	HX711 Load Cell Amplifier	1	150 – 300 BDT	1 – 3 USD
4	DHT22 Temperature & Humidity Sensor	1	300 – 600 BDT	3 – 5 USD
5	Buzzer Module	1	50 – 100 BDT	<1 USD
6	Servo Motor (Optional Bed Tilt)	1	400 – 800 BDT	3 – 7 USD
7	Power Supply (5V/12V Adapter)	1	300 – 600 BDT	3 – 5 USD
8	Wires, Breadboard, Miscellaneous	-	300 – 500 BDT	3 – 4 USD

💵 **Total Estimated Cost**
Minimum Cost: ~2900 BDT (~25 USD)
Maximum Cost: ~5100 BDT (~45 USD)

▧ Chapter 14: Conclusion

The IoT-Based Smart Patient Turning and Bed Monitoring System provides an effective and low-cost solution for continuous monitoring of bedridden patients. The system successfully integrates multiple sensors such as pressure sensors and DHT22 to track patient posture, body pressure distribution, temperature, and humidity conditions in real time.

By using the ESP32 microcontroller and IoT cloud integration, the system enables remote monitoring through mobile applications, allowing caregivers and medical staff to receive instant alerts when abnormal conditions are detected. This helps in reducing the risk of pressure ulcers (bed sores), which are a common problem among immobile patients.

The proposed system reduces the dependency on manual observation and ensures timely intervention through automated alerts such as buzzer notifications and mobile push notifications. Additionally, the optional feature of automatic bed tilting further enhances patient comfort and safety.

Overall, this project demonstrates how IoT technology can be effectively applied in healthcare to improve patient care, increase efficiency, and reduce caregiver workload. It is a scalable and cost-effective solution that can be further enhanced with AI-based prediction and hospital-level integration in the future.